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Studierichting: Informatica

Oefeningenreeks 4A nr. 18.4

$$\begin{aligned}\int x^3 \cos x dx &= \int x^3 d(\sin x) = x^3 \sin x - \int \sin x dx^3 \\&= x^3 \sin x - 3 \int x^2 \sin x dx = x^3 \sin x + 3 \int x^2 d(\cos x) \\&= x^3 \sin x + 3x^2 \cos x - 6 \int x d(\sin x) \\&= x^3 \sin x + 3x^2 \cos x - 6x \sin x + 6 \int \sin x dx \\&= x^3 \sin x + 3x^2 \cos x - 6x \sin x - 6 \cos x + c \\&= (x^3 - 6x) \sin x + (3x^2 - 6) \cos x + c\end{aligned}$$

References